Linear simulations, fixed glmnet paths and honest−CV adaptive lassos

Entries are mean (standard error) across replicates.

Model 1

Huang, AR(1) rho = 0.5; n = 100, p = 200, 15 nonzero; sigma = 1.5

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	1.2652 (0.0357)	0.5062 (0.0145)	29.6950 (0.5813)	0.9233 (0.0029)	0.9783 (0.0025)	0.0812 (0.0031)
suni2D_lambda_alpha	1.3005 (0.0364)	0.5132 (0.0150)	31.9600 (0.6970)	0.9119 (0.0035)	0.9777 (0.0024)	0.0935 (0.0038)
suni2D_sum_rate	1.2982 (0.0365)	0.5127 (0.0151)	31.8500 (0.6953)	0.9125 (0.0035)	0.9783 (0.0024)	0.0928 (0.0037)
uniLasso	1.5295 (0.0366)	0.4255 (0.0119)	20.3950 (0.3866)	0.9628 (0.0016)	0.9317 (0.0043)	0.0347 (0.0019)
lasso	1.3392 (0.0422)	0.5251 (0.0171)	33.4300 (1.0169)	0.9043 (0.0050)	0.9763 (0.0025)	0.1015 (0.0055)
adalasso_uni	1.3782 (0.0343)	0.4354 (0.0113)	23.1900 (0.5880)	0.9489 (0.0026)	0.9323 (0.0042)	0.0498 (0.0030)
adalasso_lasso	1.4942 (0.0480)	0.4060 (0.0171)	16.5800 (0.3483)	0.9793 (0.0016)	0.9147 (0.0044)	0.0155 (0.0018)

Model 2

Huang, AR(1) rho = 0.95; n = 100, p = 200, 15 nonzero; sigma = 1.5

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	5.8753 (0.1957)	0.2316 (0.0092)	16.5150 (0.1674)	0.9804 (0.0009)	0.9197 (0.0038)	0.0147 (0.0009)
suni2D_lambda_alpha	5.7444 (0.1938)	0.2389 (0.0095)	17.5650 (0.2815)	0.9748 (0.0015)	0.9177 (0.0038)	0.0205 (0.0015)
suni2D_sum_rate	5.7490 (0.1932)	0.2391 (0.0095)	17.6250 (0.2863)	0.9745 (0.0015)	0.9177 (0.0038)	0.0209 (0.0015)
uniLasso	5.8899 (0.1907)	0.1884 (0.0073)	14.9600 (0.1697)	0.9881 (0.0009)	0.9197 (0.0040)	0.0063 (0.0009)
lasso	6.0855 (0.2166)	0.2624 (0.0120)	19.8450 (0.7725)	0.9638 (0.0039)	0.9200 (0.0038)	0.0327 (0.0042)
adalasso_uni	5.8830 (0.1910)	0.1938 (0.0079)	15.5500 (0.2540)	0.9852 (0.0013)	0.9213 (0.0038)	0.0094 (0.0013)
adalasso_lasso	9.8144 (0.3932)	0.3173 (0.0153)	14.3950 (0.3054)	0.9743 (0.0014)	0.8090 (0.0060)	0.0122 (0.0015)

Model 3

Huang, 3 groups of 5 near−collinear signals; n = 100, p = 200; sigma = 1.5

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	23.5581 (0.5531)	0.3500 (0.0097)	24.6950 (0.6277)	0.9240 (0.0030)	0.8163 (0.0061)	0.0673 (0.0033)
suni2D_lambda_alpha	23.6294 (0.5555)	0.3653 (0.0108)	28.7750 (0.9024)	0.9032 (0.0044)	0.8137 (0.0063)	0.0896 (0.0048)
suni2D_sum_rate	23.2390 (0.5499)	0.3578 (0.0109)	28.0850 (0.9039)	0.9070 (0.0045)	0.8163 (0.0062)	0.0856 (0.0048)
uniLasso	26.1396 (0.6338)	0.2076 (0.0059)	15.2050 (0.2744)	0.9721 (0.0012)	0.8207 (0.0062)	0.0156 (0.0013)
lasso	24.1895 (0.5776)	0.3649 (0.0111)	28.9000 (1.0677)	0.9032 (0.0053)	0.8177 (0.0065)	0.0899 (0.0057)
adalasso_uni	23.9264 (0.5856)	0.1945 (0.0058)	15.7550 (0.3757)	0.9695 (0.0018)	0.8217 (0.0061)	0.0185 (0.0019)
adalasso_lasso	56.7092 (1.6229)	0.3550 (0.0123)	10.8850 (0.5476)	0.9479 (0.0023)	0.5160 (0.0083)	0.0170 (0.0026)

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Model 4

uniLasso, equicorrelated rho = 0.5; n = 300, p = 1000, 30 nonzero ~ U(0, 1); sigma = 0.1

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	0.0396 (0.0006)	5.1094 (0.0807)	111.2400 (0.4907)	0.9172 (0.0005)	0.9745 (0.0018)	0.0845 (0.0005)
suni2D_lambda_alpha	0.0123 (0.0002)	0.6438 (0.0105)	119.2100 (0.6589)	0.9099 (0.0007)	0.9853 (0.0014)	0.0924 (0.0007)
suni2D_sum_rate	0.0123 (0.0002)	0.6442 (0.0105)	119.2200 (0.6597)	0.9099 (0.0007)	0.9853 (0.0014)	0.0924 (0.0007)
uniLasso	0.0140 (0.0002)	1.2189 (0.0169)	123.8000 (0.6414)	0.9052 (0.0006)	0.9828 (0.0015)	0.0972 (0.0007)
lasso	0.0155 (0.0002)	0.8325 (0.0128)	140.6650 (1.8621)	0.8883 (0.0018)	0.9835 (0.0015)	0.1146 (0.0019)
adalasso_uni	0.0157 (0.0002)	0.8527 (0.0129)	144.2800 (1.6764)	0.8847 (0.0017)	0.9833 (0.0015)	0.1183 (0.0017)
adalasso_lasso	0.0029 (0.0001)	0.1574 (0.0037)	32.1450 (0.2760)	0.9961 (0.0003)	0.9715 (0.0020)	0.0031 (0.0003)

Model 5

uniLasso home court, AR(1) rho = 0.8; n = 100, p = 30, 20% nonzero; SNR = 1

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	7.4253 (0.3863)	0.1448 (0.0056)	10.7000 (0.2641)	0.7405 (0.0081)	0.7450 (0.0128)	0.2596 (0.0101)
suni2D_lambda_alpha	7.4423 (0.3899)	0.1451 (0.0056)	10.6750 (0.2665)	0.7413 (0.0082)	0.7450 (0.0128)	0.2585 (0.0102)
suni2D_sum_rate	7.4348 (0.3892)	0.1451 (0.0056)	10.6750 (0.2652)	0.7413 (0.0081)	0.7450 (0.0128)	0.2585 (0.0101)
uniLasso	6.3710 (0.3080)	0.1425 (0.0052)	6.9250 (0.1483)	0.8215 (0.0051)	0.6308 (0.0133)	0.1308 (0.0053)
lasso	7.6116 (0.4331)	0.1448 (0.0054)	10.8450 (0.2828)	0.7348 (0.0085)	0.7425 (0.0130)	0.2662 (0.0107)
adalasso_uni	8.3314 (0.5981)	0.1405 (0.0050)	9.8550 (0.2861)	0.7562 (0.0086)	0.7142 (0.0129)	0.2321 (0.0108)
adalasso_lasso	10.7692 (0.6723)	0.1845 (0.0079)	6.5950 (0.2700)	0.8075 (0.0068)	0.5692 (0.0151)	0.1325 (0.0092)